

MISSION 1: System Health & Monitoring

Where Am I and Is Anything on Fire?

First Look at the System

Command	Key Flags	What It Does / Key Output
uptime	(none)	Server uptime + load averages. Load > core count = overloaded. Check user count for unexpected sessions.
free -h	-h -m -s N	RAM usage. Ignore 'free' column. Check 'available' -- if <10% of total, trouble. High swap = out of memory.
df -h	-h -T -i	Disk usage. Use% >90% = danger. 100% = apps crash. Check / and /var first.
top	P M 1 k q	Live process monitor. Watch %CPU, %MEM, RES. 'wa' in header >10% = disk bottleneck.
htop	F5 F9 / F6	Colorful top. Green=user, Red=kernel, Blue=low-pri. Install: apt install htop
vmstat 1 5	interval count	CPU/mem/I/O snapshot. wa=disk, si/so=swap, r>cores=overloaded, b>0=IO blocked.
iostat -x 1 3	-x (extended)	Disk I/O detail. await <10ms=good, >50ms=slow. %util=100%=saturated.

TIP: Golden rule: high wa = disk problem, high us = app problem, high sy = kernel problem. If si/so active, add RAM.

History & Logs

Command	Key Flags	What It Does / Key Output
dmesg -T	-T grep -i	Kernel messages. Look for: OOM killer, I/O errors, segfaults, network link changes.
journalctl	-u svc -f -p err -b	Systemd master log. -p err -b = all errors since boot. -f = follow live. -u = filter by service.
sar	-u -r -n DEV -d	Historical stats (DVR for server). Install sysstat. Records every 10min. Rewind to find 3AM crashes.

Services & Processes

Command	Key Flags	What It Does / Key Output
systemctl	status restart enable	Control services. list-units --failed = find all broken. daemon-reload after editing .service files.
ps aux	--sort=-%mem -%cpu	All processes. a=all users, u=details, x=daemons. STAT: S=sleep, R=run, Z=zombie, D=IO wait.

TIP: Day 1 combo: systemctl list-units --failed + journalctl -p err -b = instant picture of everything broken.

MISSION 2: Logs, Files & Network

The App Is Slow -- Find the Problem

Log Analysis & File Operations

Command	Key Flags	What It Does / Key Output
grep	<code>-i -c -n -A3 -B1 -r</code>	Search text. <code>-c</code> =count, <code>-n</code> =line nums, <code>-A/B</code> =context, <code>-r</code> =recursive. Use <code>\ </code> for OR patterns.
sed	<code>-n 's/old/new/' -i</code>	Stream edit. <code>-i</code> =in-place (dry run first!). <code>'s///g'</code> =global replace. <code>'/pat/d'</code> =delete lines.
awk	<code>{print \$1}' -F':'</code>	Column extraction. <code>\$0</code> =whole line. <code> sort uniq -c sort -rn</code> = frequency analysis.
find	<code>-name -type -mtime -size</code>	Find files. <code>-mtime +30</code> =older 30d. <code>-size +50M</code> =big. ALWAYS <code>-print</code> before <code>-delete</code> !
du -sh	<code>-s -h sort -rh</code>	Directory sizes. Pipe to <code>sort -rh head -10</code> for biggest dirs.
xargs	<code>-I {} -n1 -P4</code>	Pipe to commands. <code>find xargs wc -l -I {}</code> for placeholder. <code>-P4</code> for parallel.

TIP: `awk | sort | uniq -c | sort -rn` -- the most powerful one-liner. Turns any log into a frequency table.

Network Diagnostics

Command	Key Flags	What It Does / Key Output
curl	<code>-I -s -o /dev/null -w</code>	HTTP client. <code>-w</code> shows timing (DNS, connect, total). <code>-I</code> =headers only. <code>-s</code> =silent.
ss -tuln	<code>-t -u -l -n -p</code>	Socket stats. <code>0.0.0.0:80</code> =all interfaces. <code>127.0.0.1</code> =localhost only. <code>-p</code> shows process.
netstat -tuln	(same as ss)	Older version of ss. Install: <code>apt install net-tools</code> . Know both.
dig +short	<code>@8.8.8.8 +trace</code>	DNS lookup. Empty = DNS broken. Check <code>/etc/resolv.conf</code> . Half of issues are DNS.
ip addr/route	<code>addr show route show</code>	IP config. 'state UP/DOWN' for interfaces. 'default via' = gateway.
ifconfig	(deprecated)	Legacy IP config. Same as <code>ip addr</code> . Still in many docs.
mtr	<code>--report -c 10</code>	Traceroute + ping. <code>Loss% >5%</code> = problem hop. <code>Avg jump</code> = slow link.
tcpdump	<code>-i eth0 port X -c N</code>	Packet capture. <code>-c</code> =limit. <code>-w</code> =save for Wireshark.
nmap	<code>-p ports -p- -sV</code>	Port scanner. open/closed/filtered. <code>-sV</code> =version detection.

TIP: `curl -w 'DNS:%{time_namelookup} Connect:%{time_connect} Total:%{time_total}' -- pinpoints where slowness is.`

MISSION 3: Security, Users & Processes

Lock It Down

User & Permission Management

Command	Key Flags	What It Does / Key Output
id	id username	Show user/group IDs. Root = uid=0. Groups determine file access.
usermod -aG	-aG grp user -L -U	Modify user. ALWAYS use -aG (append). Without -a = REPLACE all groups (disaster!).
groupmod	-n newname old	Rename group. Members stay intact.
visudo	(safe editor)	Edit /etc/sudoers with syntax checking. NEVER use vim/nano on sudoers directly.
chmod	600 640 755 777	Change perms. r=4 w=2 x=1. 640=owner rw, group r. 777=NEVER in production.
chown	user:group file	Change ownership. Example: chown root:sre-team config.env
umask	022 027	Default perms for new files. Umask removes perms. 027 = new files get 640.

TIP: Permission cheat: 600=private, 640=team-read, 755=public-exec, 777=NEVER. Always chmod BEFORE sharing configs.

SSH & Firewall

Command	Key Flags	What It Does / Key Output
sshd config	PermitRootLogin etc.	Check /etc/ssh/sshd_config. Set PermitRootLogin + PasswordAuth to 'no'. Add SSH key FIRST!
iptables	-L -n -v -A -j	Firewall rules. -j ACCEPT/DROP/REJECT. Rules are top-to-bottom -- ORDER MATTERS.

Process Control

Command	Key Flags	What It Does / Key Output
kill	-15 -9 -1	SIGTERM(-15)=graceful. SIGKILL(-9)=force, last resort. SIGHUP(-1)=reload config.
pkill / pgrep	-la -f -u user	Kill/find by name. -f matches full command line. -u filters by user.
nice / renice	-n VAL (-20 to 19)	Process priority. nice=at start, renice -p PID=while running. Lower = higher priority.
lsof	-p PID -i :port /path	Open files/ports. 'lsof -i :80' = what uses port 80. Solves 'port in use' instantly.
strace	-p PID -e trace -c -f	-c = summary table (hidden gem). Shows call counts, errors, time. Reveals hidden failures.

TIP: strace -c shows EXACTLY where a process spends time. The 'errors' column reveals failures the app doesn't log.

MISSION 4: Deploy, Backup & Automate

Never Do This Manually Again

File Operations & Deployment

Command	Key Flags	What It Does / Key Output
tar	-czf -xzf -tzf -C	Archive/compress. czf=Create+gZip+File. -C=extract to dir. -v=verbose.
rsync	-avz --dry-run --delete	Smart sync. -a=preserves all. --dry-run FIRST! Trailing slash: /src/ vs /src matters!
tmux	new -s attach -t	Terminal multiplexer. SSH dies? tmux survives. Ctrl+B D=detach. %=split vertical.

TIP: rsync trailing slash: /src/ copies CONTENTS. /src (no slash) copies the DIRECTORY itself. This trips up everyone.

Automation

Command	Key Flags	What It Does / Key Output
crontab	-e -l	Schedule jobs. 0 2 * * * = 2AM daily. */15=every 15min. ALWAYS add >> log 2>&1 for logging.
logrotate	daily rotate N compress	Auto-manage log files. -f=force now. Config in /etc/logrotate.d/. Prevents disks filling up.

Cron Quick Reference

Expression	Meaning
0 2 * * *	Every day at 2:00 AM
*/15 * * * *	Every 15 minutes
0 9 * * 1-5	9 AM, Monday through Friday
0 0 1 * *	Midnight on 1st of every month

Disk & Storage

Command	Key Flags	What It Does / Key Output
lsblk	(none)	List block devices. NAME, SIZE, TYPE (disk/part/lvm), MOUNTPOINT (empty=not mounted).
fdisk -l	-l (list)	List all partitions. Use fdisk for <2TB, parted for larger.
mount/umount	mount dev /path	Make partitions accessible. Add to /etc/fstab for persistence across reboots.

Deep System Tools

Command	Key Flags	What It Does / Key Output
chroot	/mnt/recovery /bin/bash	Change root dir. System recovery tool. Docker uses this concept under the hood.
ldd	/usr/bin/nginx	Show shared libraries. 'not found' = missing dependency causing app startup failure.
udevadm	info --query=all	Device event manager. Mostly bare-metal/embedded DevOps.

TIP: When app says 'error loading shared libraries', run ldd on the binary. It shows exactly which library is missing.

COMPLETE COMMAND INDEX

All 50 commands at a glance -- sorted by mission

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4	top	System	29	usermod	Users
5	htop	System	30	groupmod	Users
6	vmstat	System	31	visudo	Users
7	iostat	System	32	chmod/chown	Perms
8	dmesg	History	33	umask	Perms
9	journalctl	History	34	sshd	SSH
10	sar	History	35	iptables	Firewall
11	systemctl	Services	36	kill	Process
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14	sed	Logs	39	lsof	Process
15	awk	Logs	40	strace	Process
16	find	Files	41	tar	Deploy
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19	curl/wget	Network	44	crontab	Automate
20	ss	Network	45	logrotate	Automate
21	netstat	Network	46	lsblk	Storage
22	dig	Network	47	fdisk	Storage
23	ip addr	Network	48	mount	Storage
24	ifconfig	Network	49	chroot	Deep
25	mtr	Network	50	ldd	Deep

Emergency Quick Reference

Situation	Run This
Server slow	top > check wa% > iostat -x 1 3 > free -h
Disk full	df -h > du -sh /* sort -rh head > find / -size +100M
App crashed	systemctl status app > journalctl -u app -n 50 > dmesg tail
Can't connect	dig host +short > ss -tuln > iptables -L -n > curl -I url
Port in use	lsof -i :PORT > kill PID (or kill -9 PID)
Out of memory	free -h > ps aux --sort=-%mem head > dmesg grep oom
Who did what	journalctl -p err -b > last > grep -i error /var/log/syslog

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